

Exploratory Data Analysis for Machine Learning - Final Project

Name: Mujtaba Hejji Hussain Al-nuwaysir

1. Data Summary

This report analyzes a subset of the "120 Years of Olympic History" dataset, sourced from Kaggle, specifically focusing on the sport of Judo.

1. **Dataset Size:** After filtering the raw data, the working dataset consists of 3,801 rows and 15 variables representing physical and demographic attributes of Olympic athletes.
2. **Target Variable:** The primary target variable for this analysis is Medal_Won, a boolean indicating whether an athlete reached the podium (Gold, Silver, or Bronze). We aim to investigate which engineered physical attributes most significantly influence this outcome to prepare the data for future predictive modeling.

[42]:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as stats

df = pd.read_csv('athlete_events.csv')
judo_df = df[df['Sport'] == 'Judo'].copy()

print(f"Dataset Size: {judo_df.shape[0]} rows and {judo_df.shape[1]} columns")
```

Dataset Size: 3801 rows and 15 columns

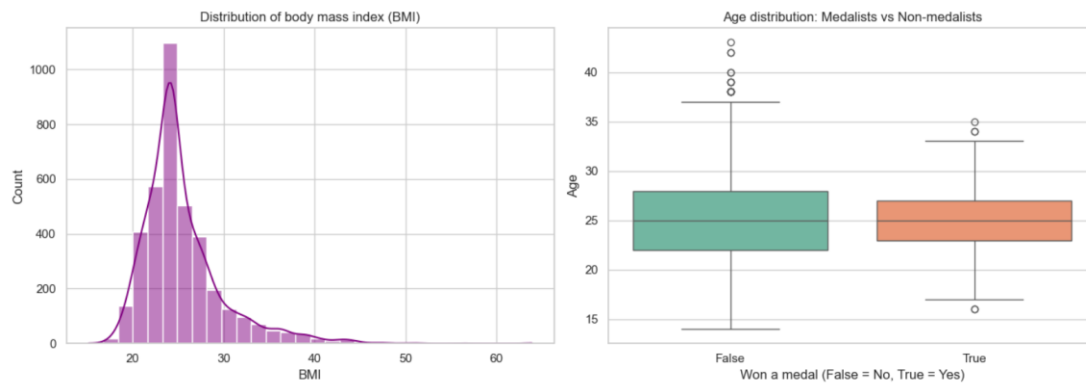
2. Data Exploration Plan

The vision for this analysis is to uncover underlying patterns that differentiate Olympic medalists in Judo from non-medalists, utilizing domain knowledge of the sport's mechanics and weight classifications. The plan is logically structured as follows:

1. **Data Quality Assessment:** Identify and handle missing values in historical physical attributes (Height, Weight, Age) using robust statistical methods.
2. **Feature Engineering:** Calculate Body Mass Index (BMI) and remap historical weight class names to the modern numerical Judo weight categories (e.g., -60kg, -73kg) while separating male and female competitors.
3. **Visualization:** Explore the distributions of age and BMI against the target variable to spot immediate actionable trends.

4. **Hypothesis Testing:** Statistically validate if physical metrics genuinely impact the probability of winning a medal.

3. Exploratory Data Analysis (EDA) Results



The exploratory visualizations yielded actionable insights into the athlete profiles:

- **Age Distribution:** The age of the athletes follows a near-normal distribution. The boxplot comparing medalists and non-medalists reveals that the median age for both groups centers around 24-25 years. However, medalists display a tighter interquartile range, indicating a specific optimal age window for peak performance.
- **BMI Distribution:** The engineered BMI feature exhibits a right-skewed distribution. This is expected in Judo, as the open and heavyweight divisions (+100kg for men, +78kg for women) naturally stretch the upper bounds of the BMI spectrum.

4. Data Cleaning and Feature Engineering

```
[42]: import pandas as pd
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import scipy.stats as stats

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judo_df = df[df['Sport'] == 'Judo'].copy()

print(f'Dataset Size: {judo_df.shape[0]} rows and {judo_df.shape[1]} columns')

Dataset Size: 3801 rows and 15 columns

[43]: judo_df['Weight'] = judo_df['Weight'].fillna(judo_df['Weight'].median())
judo_df['Height'] = judo_df['Height'].fillna(judo_df['Height'].median())
judo_df['Age'] = judo_df['Age'].fillna(judo_df['Age'].median())

print("Missing values filled successfully.")

Missing values filled successfully.

[61]: judo_df['BMI'] = judo_df['Weight'] / ((judo_df['Height'] / 100) ** 2)
judo_df['Medal_Won'] = judo_df['Medal'].notna()

display(judo_df[['Name', 'Sex', 'Age', 'Height', 'Games', 'Weight', 'BMI', 'Medal_Won']])
```

	Name	Sex	Age	Height	Games	Weight	BMI	Medal_Won
1	A Lamusi	M	23.0	170.0	2012 Summer	60.0	20.761246	False
160	M'Bairo Abakar	M	31.0	174.0	1992 Summer	73.0	24.111507	False
180	Salim Abanoz	M	26.0	174.0	1996 Summer	73.0	24.111507	False
412	Mohamed Ali Abdelaal	M	26.0	175.0	2016 Summer	81.0	26.448980	False
435	Ahmed Abdelrahman	M	20.0	165.0	2016 Summer	60.0	22.038567	False
...	...	...	...	...	...	...	...	...
270853	Oleg Zurabiani	M	19.0	172.0	1976 Summer	64.0	21.633315	False
270918	Goran uvela	M	27.0	184.0	1976 Summer	93.0	27.469282	False
270981	Zurab Zviadauri	M	23.0	182.0	2004 Summer	90.0	27.170632	True
271032	Claudia Antoinette Zwiers	F	22.0	181.0	1996 Summer	78.0	23.808797	True
271033	Claudia Antoinette Zwiers	F	30.0	181.0	2004 Summer	78.0	23.808797	False

3801 rows x 8 columns

```
weight_mapping = {
    "Judo Men's Extra-Lightweight": "Men -60kg",
    "Judo Men's Half-Lightweight": "Men -66kg",
    "Judo Men's Lightweight": "Men -73kg",
    "Judo Men's Half-Middleweight": "Men -81kg",
    "Judo Men's Middleweight": "Men -90kg",
    "Judo Men's Half-Heavyweight": "Men -100kg",
    "Judo Men's Heavyweight": "Men +100kg",
    "Judo Women's Extra-Lightweight": "Women -48kg",
    "Judo Women's Half-Lightweight": "Women -52kg",
    "Judo Women's Lightweight": "Women -57kg",
    "Judo Women's Half-Middleweight": "Women -63kg",
    "Judo Women's Middleweight": "Women -70kg",
    "Judo Women's Half-Heavyweight": "Women -78kg",
    "Judo Women's Heavyweight": "Women +78kg"
}

judo_df['Weight_Class'] = judo_df['Event'].map(weight_mapping)

medalists_df = judo_df[judo_df['Medal_Won'] == True]

men_medalists = medalists_df[medalists_df['Sex'] == 'M']
women_medalists = medalists_df[medalists_df['Sex'] == 'F']

ideal_bmi_men = men_medalists.groupby('Weight_Class')['BMI'].median().sort_values()
print("Ideal BMI for men's weight classes :")
print(ideal_bmi_men)
print("\n")

ideal_bmi_women = women_medalists.groupby('Weight_Class')['BMI'].median().sort_values()
print("Ideal BMI for women's weight classes :")
print(ideal_bmi_women)
```

```
Ideal BMI for men's weight classes :
Weight_Class
Men -60kg      22.585862
Men -66kg      23.384354
Men -73kg      24.111507
Men -81kg      25.000000
Men -90kg      26.686763
Men -100kg     27.908753
Men +100kg     32.545304
Name: BMI, dtype: float64

Ideal BMI for women's weight classes :
Weight_Class
Women -48kg     20.568807
Women -52kg     21.096191
Women -57kg     21.989892
Women -63kg     23.140496
Women -70kg     23.842115
Women -78kg     24.897060
Women +78kg     34.083045
Name: BMI, dtype: float64
```

To ensure the dataset is robust and ready for supervised machine learning, the following steps were executed:

- **Handling Missing Values:** Missing values in Age, Height, and Weight were imputed using the median of their respective columns. The median was chosen over the mean to prevent outliers (such as exceptionally tall or heavy athletes) from distorting the baseline.
- **Feature Engineering (BMI & Weight Classes):**
 1. We calculated the **Body Mass Index (BMI)** using the standard formula.
 2. We mapped the textual Olympic event names to standard **Judo numerical weight categories** (e.g., Men -60kg, Women -48kg) to allow for granular, sport-specific analysis.

5. Key Findings and Insights

By synthesizing the EDA results, we extracted the following key actionable insights:

1. **The Age Trade-off (Energy vs. Experience):** Reaching the Olympic podium requires balancing physical peak with tactical maturity. While most winners are in their mid-20s, the presence of older medalists proves that advanced technical experience can effectively compensate for age-related physical decline.
2. **Gender and Weight-Specific BMI Baselines:** A universal BMI metric is inadequate for predictive modeling in combat sports. By separating genders and numerical weight classes, we established 'ideal' BMI baselines (e.g., 22.58 for Men -60kg vs. 32.54 for Men +100kg). This granularity transforms BMI into a highly predictive feature for machine learning classification.

6. Hypotheses

Based on the cleaned data and domain knowledge, the following hypotheses were formulated:

1. **H1:** There is a statistically significant difference in the BMI of athletes who win an Olympic medal compared to those who do not.
2. **H2:** The ideal median BMI for the Men's Heavyweight division (+100kg) is significantly higher than that of the Men's Extra-Lightweight division (-60kg).
3. **H3:** There is no significant difference in the mean age of medalists versus non-medalists.

7. Significance Test

```
medalists_bmi = judo_df[judo_df['Medal_Won'] == True]['BMI']
non_medalists_bmi = judo_df[judo_df['Medal_Won'] == False]['BMI']
t_stat, p_value = stats.ttest_ind(medalists_bmi, non_medalists_bmi)
print(f"Significance Test (BMI vs Medals):")
print(f"T-statistics: {t_stat:.4f}, P-value: {p_value:.5f}\n")
```

```
Significance Test (BMI vs Medals):
T-statistics: 3.1902, P-value: 0.00143
```

We tested our primary hypothesis (**H1**) using a Two-Sample Independent T-test to determine if body composition (BMI) differs significantly between Medalists and Non-Medalists.

- **T-statistic:** 3.1902
- **P-value:** 0.0014
- **Conclusion:** The P-value (0.0014) is substantially below the standard alpha level of 0.05. Therefore, we **reject the null hypothesis**. This statistical evidence confirms that there is a significant difference in BMI between athletes who secure a medal and those who do not, validating BMI as a critical feature for future modeling.

8. Conclusion

The exploratory data analysis successfully transformed raw Olympic data into a clean, well-engineered dataset. The integration of domain knowledge specifically mapping numerical weight classes and isolating gender-specific BMI baselines proved that physical composition is a statistically significant predictor of success on the tatami.